

Homework (Carolina Reaper)

- Q1.** Factor the trinomial $x^2 + 5x + 6$ to determine the dimensions of a painting canvas.
(A) $(x + 3)(x + 2)$
(B) $(x + 4)(x + 1)$
(C) $(x + 2)(x + 4)$
(D) $(x + 1)(x + 6)$
(E) $(x + 6)(x + 1)$
- Q2.** The frequency of a musical note is given by $3x^2 - 12x - 15$. Factor this expression.
(A) $3(x + 1)(x - 5)$
(B) $3(x - 1)(x + 5)$
(C) $3(x - 5)(x + 1)$
(D) $3(x + 5)(x - 1)$
(E) $3(x - 3)(x + 3)$
- Q3.** Factor $x^2 - 7x + 12$ to find the rhythm pattern of a song.
(A) $(x - 4)(x - 3)$
(B) $(x - 3)(x - 4)$
(C) $(x + 4)(x + 3)$
(D) $(x + 3)(x + 4)$
(E) $(x + 2)(x + 5)$
- Q4.** The area of a sculpture is given by $2x^2 + 5x + 2$. Factor this expression.
(A) $(2x + 1)(x + 2)$
(B) $(2x + 2)(x + 1)$
(C) $(2x - 1)(x + 2)$
(D) $(2x - 2)(x - 1)$
(E) $(2x + 1)(x - 2)$
- Q5.** The length of a guitar string is given by $x^2 - 9x + 20$. Factor this expression.
(A) $(x - 5)(x - 4)$
(B) $(x + 5)(x + 4)$
(C) $(x - 4)(x - 5)$
(D) $(x + 4)(x + 5)$
(E) $(x - 2)(x - 10)$
- Q6.** The volume of a speaker is given by $x^2 + 2x - 15$. Factor this expression.
(A) $(x + 5)(x - 3)$
(B) $(x + 3)(x - 5)$
(C) $(x - 5)(x + 3)$
(D) $(x - 3)(x + 5)$
(E) $(x + 4)(x - 4)$
- Q7.** Factor the trinomial $3x^2 + 2x - 1$ to determine the beats per minute of a song.
(A) $3x^2 + 3x - x - 1$
(B) $3x^2 - 3x + 5x - 1$
(C) $(3x - 1)(x + 1)$
(D) $(3x + 1)(x - 1)$
(E) $(x + 1)(3x - 1)$
- Q8.** The resonance of a violin string is given by $x^2 - 4x - 21$. Factor this expression.
(A) $(x - 7)(x + 3)$
(B) $(x + 7)(x - 3)$
(C) $(x + 3)(x - 7)$
(D) $(x - 3)(x + 7)$
(E) $(x - 4)(x - 6)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $2x^2 + 11x + 12$ and explain how it relates to the frequency of a musical instrument. Provide your answer in 5 lines or less.
- Q10.** A painter is creating a mural on a canvas with dimensions given by the factored form of $x^2 - 5x - 14$. Determine the factors and describe how they relate to the size of the mural. Provide your answer in 5 lines or less.

Chef's Tips

- Check for factoring errors
- Ensure all work is shown
- Use only approved calculator models